BLOCKCHAIN TECHNOLOGY-BCS613A QUESTION BANK

Module-1

1.Explain how blockchain accumulates blocks?

2.With a neat diagram, Explain the CAP Theorem. How does it apply to the design and operation of blockchain networks?

3. Explain the generic elements of a blockchain along with a diagram?

4.With a neat diagram, discuss the Byzantine Generals Problem. How does the blockchain technology overcomes this problem?

5. What is block chain technology? Discuss the different tiers in blockchain technology? What are the different types in blockchain technology?

6. Write a note on features and applications of blockchain technology?

7.Explain benefits and limitation of blockchain technology?

8.What is Consensus mechanism? What are the different consensus used in Blockchain technology?

9.Write a note on Practical Byzantine Fault tolerance [PBFT]?

Module-2

1.What is decentralization in blockchain technology? What are the different methods of decentralisation?

2.Write a note on Decentralized ecosystem in blockchain with a neat diagram?

3. Explain the following Pertinent terminologies?

1. Smart Contracts
2. Autonomous Agents (AAs)
3. Decentralized Organizations (DOs)
4. Decentralized autonomous organization (DAOs)
5. Decentralized Autonomous Corporations (DACs)

4.What is decentralised APPS [DAPPS]? Mention its types?

5.Discuss the requirements and operations of decentralised applications?

6.Write a note on how the design of DAPPS differs from Traditional Application architecture?

7.Write a note on DAPP examples and platforms used for decentralization?

8.Write a note on innovative trends in Blockchain Technology?

9.With a neat diagram, explain Taxonomy of Cryptographic primitives?

10.What are the three security properties of hash function?

11.Using Merkel Damagard construction, explain SHA-256 with a neat sketch?

12.Using Sponge & Squeeze Construction model, Write a note on design on SHA-3 [KECCAK]?

13.Write a note on

i. Merkle Trees
ii. Patricia Trees

14.Explain the Working of

a. RSA algorithm

b. AES algorithm

c. ECDSA-Elliptic curve digital signature algorithm

15.Write a note on **Point addition, Point multiplication, Point doubling** with a neat diagram in elliptic curve cryptography?

16. What is signatures? Write a note on its types?

17.Explain the concept of ZK-Snarks?

Module-3

1. Define Bitcoin. Discuss its architecture and explain how it differs from traditional digital currencies.

2.Explain the Bitcoin transaction life cycle with a neat diagram. Highlight the major stages involved.

3.With the help of a diagram, explain the structure of a Bitcoin block and the structure of a block header.

4.What is a Genesis Block? Describe its significance in maintaining the security and continuity of the blockchain.

5.Discuss the working of the Bitcoin network. How does the network achieve consensus without a central authority?

6.What are Bitcoin wallets? Explain different types of wallets and discuss their advantages and disadvantages.

7.Describe Smart Contracts. Discuss their history, evolution, and the advantages they offer over traditional contracts.

8.What are Ricardian Contracts? Compare and contrast Ricardian Contracts and Smart Contracts.

9.Explain the concept of Oracles in blockchain. How do Smart Oracles enhance the functionality of smart contracts? Give examples.

10.Explain the step-by-step procedure to deploy a smart contract on a public blockchain platform like Ethereum.

11.What is "The DAO"? Explain its objectives, failure, and the major consequences it had on the Ethereum ecosystem.

12.Discuss the importance of Smart Contract Templates and their role in simplifying smart contract development.

Module-4

1.Explain Ethereum in detail. Discuss its architecture, key features, and how it differs from Bitcoin.

2.What are Ethereum clients? List some major Ethereum client releases and explain the purpose of using different clients.

3.Describe the components of the Ethereum technology stack. How does each layer contribute to the overall blockchain ecosystem?

4.Explain the Ethereum blockchain structure. How are blocks created, validated, and linked together?

5.What is Ether? Compare ETH and ETC, explaining the circumstances that led to the Ethereum fork.

6.Discuss the concept of Gas in Ethereum. Why is Gas important, and how does it impact transaction execution?

7.Explain the consensus mechanism used by Ethereum. Compare Ethereum's Proof of Work (PoW) and Proof of Stake (PoS) models.

8.What is the World State in Ethereum? How does it differ from the Bitcoin UTXO model?

9.Describe the types of transactions in Ethereum. How do contract creation transactions and message call transactions differ?

10.Explain the architecture and functioning of the Ethereum Virtual Machine (EVM). Why is EVM considered a critical component of Ethereum?

11.Describe the role of accounts, blocks, ether, messages, and mining in the Ethereum network. How do they interrelate?

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12.Explain the mining process in Ethereum. How are miners rewarded and what changes occurred after the Ethereum Merge (if applicable)?

13.What is Geth? Explain how Geth is used for managing Ethereum clients and wallets. Describe some common Geth commands.

Module-5

1.What is Hyperledger? Explain its objectives and describe the different projects under the Hyperledger umbrella.

2.Discuss Hyperledger as a protocol. How does it support enterprise blockchain solutions? Explain its key features.

3.What is Fabric in Hyperledger? Explain the architecture and major components of Hyperledger Fabric.

4.Explain how Hyperledger Fabric differs from public blockchains like Bitcoin and Ethereum. Why is it suitable for enterprise applications?

5.What are the main features and advantages of Hyperledger Fabric? Discuss the concepts of channels, chaincode, and endorsement policies.

6.Describe the architecture of Sawtooth Lake. How does its consensus mechanism (like PoET) differ from traditional Proof-of-Work?

7.What is Corda? Explain the design goals of Corda and how it differs from other blockchain platforms like Hyperledger Fabric and Ethereum.

8.Compare and contrast Hyperledger Fabric, Sawtooth Lake, and Corda based on architecture, consensus mechanism, and use cases.

9.Discuss real-world applications of Hyperledger Fabric and Corda in industries like finance, healthcare, and supply chain management.

10.Explain the role of smart contracts in Hyperledger Fabric. How are they different from smart contracts in Ethereum?

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